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Untitled9.ipynb - Colab

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Untitled9.ipynb

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import numpy as np

XOR dataset

X = np.array([[0,0],
[0,1],
[1,0],
[1,1]])

Y = np.array([[0],
[1],
[1],
[0]])

Initialize weights randomly

np.random.seed(0)

W1 = np.random.rand(2,2)

W2 = np.random.rand(2,1)

Sigmoid function

def sigmoid(x):

return 1/(1+np.exp(-x))

Derivative

def derivative(x):

return x*(1-x)

Training (Backpropagation)

for i in range(5000):

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Forward pass

hidden = sigmoid(np.dot(X, W1))

output = sigmoid(np.dot(hidden, W2))

Error

error = Y - output

Backward pass

d_output = error * derivative(output)

d_hidden = d_output.dot(W2.T) * derivative(hidden)

Update weights

W2 += hidden.T.dot(d_output)

W1 += X.T.dot(d_hidden)

Final Output

print("Trained Output:\n", output)

Trained Output:
[[0.05472325]
[0.89768011]
[0.89768013]
[0.13587921]]

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